

# Project Lent Week 3

## Synthetic Data Analysis

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### 1 Overview

- Fixed OAK implementation and reran synthetic datasets
- Got the individual fittings for each feature
- Researched DKL, coded and ran a GP/NN/DKL Comparison
- Researched into  $\beta$ -VAE as a DKL solution

## 2 Synthetic Data Types and Results

### 2.1 Type 0: Periodic Function with Noise

This dataset is generated using a sine wave with an added Gaussian noise term:

```
x_1 = np.arange(0, 100.2, 0.2)
x_2 = np.random.normal(loc=0.7, scale=0.1, size=len(x_1))
y = np.sin(2 * x_1) + x_2
```

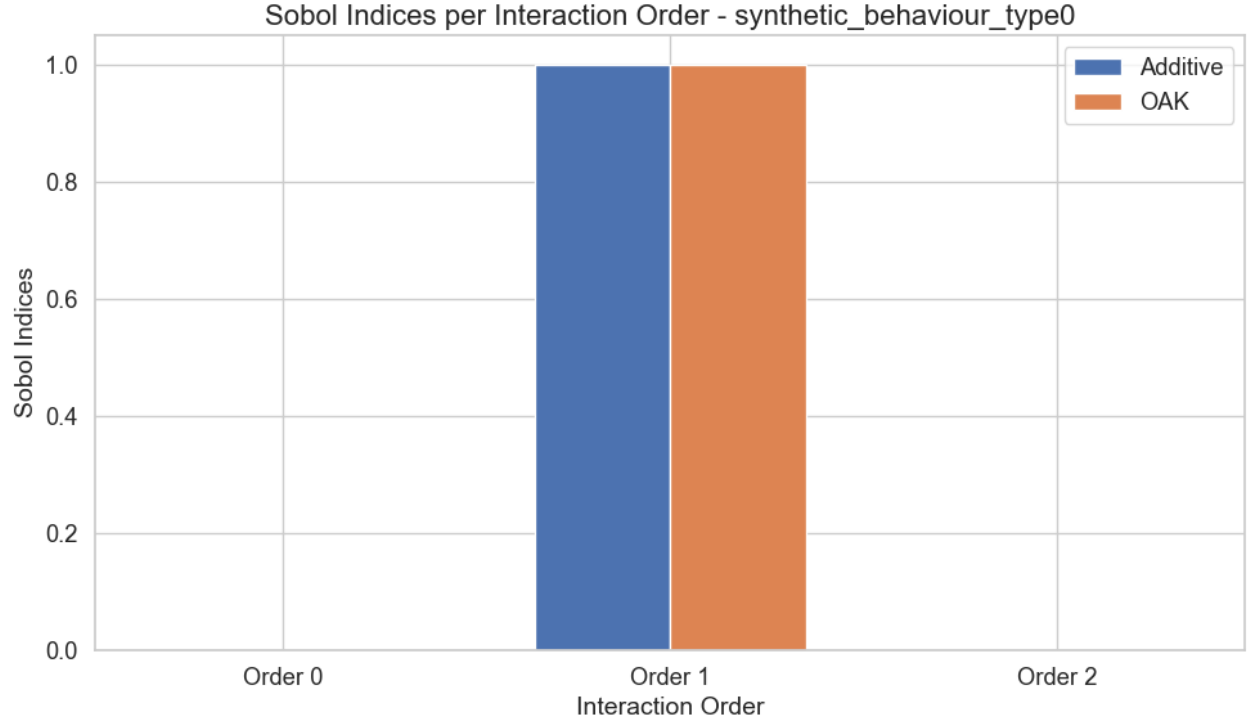


Figure 1: Sensitivity analysis for Type 0 data.

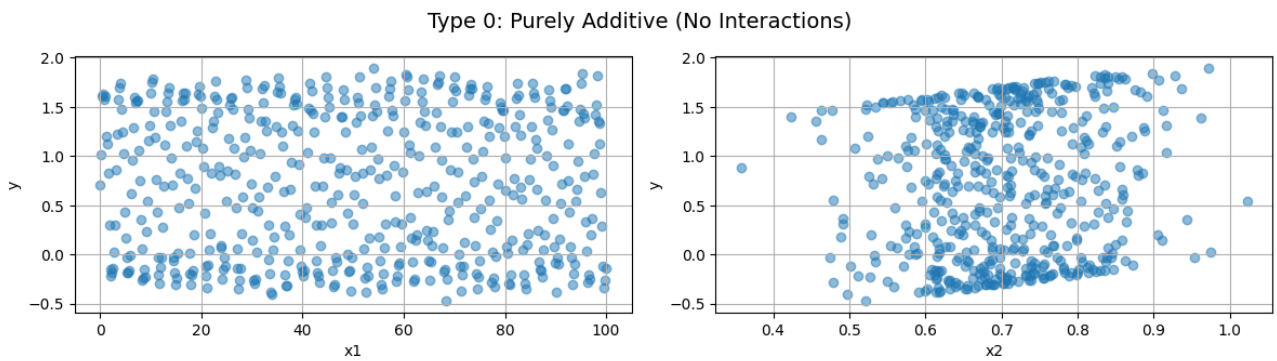


Figure 2: Marginal distributions of Type 0 data.

Type 0: Used for testing

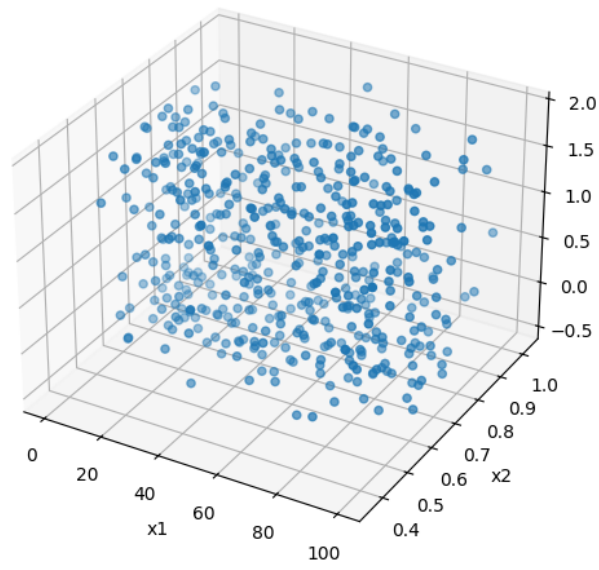


Figure 3: Joint distribution of Type 0 data.

## 2.2 Type 1: Purely Additive Function

This dataset has no interaction between features, where the response variable is a sum of independent transformations:

```
x_1 = np.random.normal(loc=0.7, scale=0.1, size=500)
x_2 = np.linspace(0, 10, len(x_1))
x_3 = np.random.uniform(-1, 1, size=len(x_1))
y = np.sin(x_1) + np.log(1 + x_2) + np.exp(x_3)
```

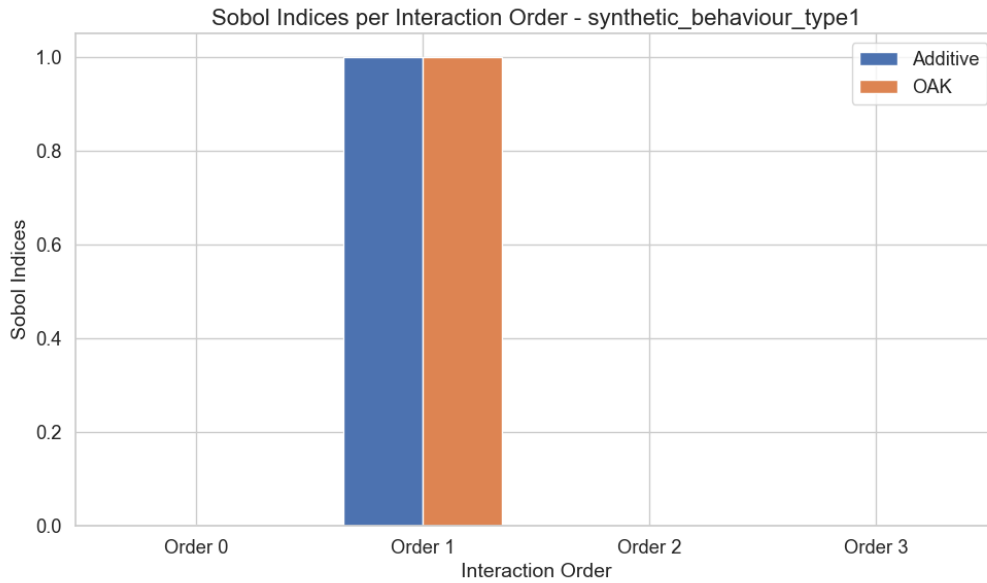


Figure 4: Sensitivity analysis for Type 1 data.

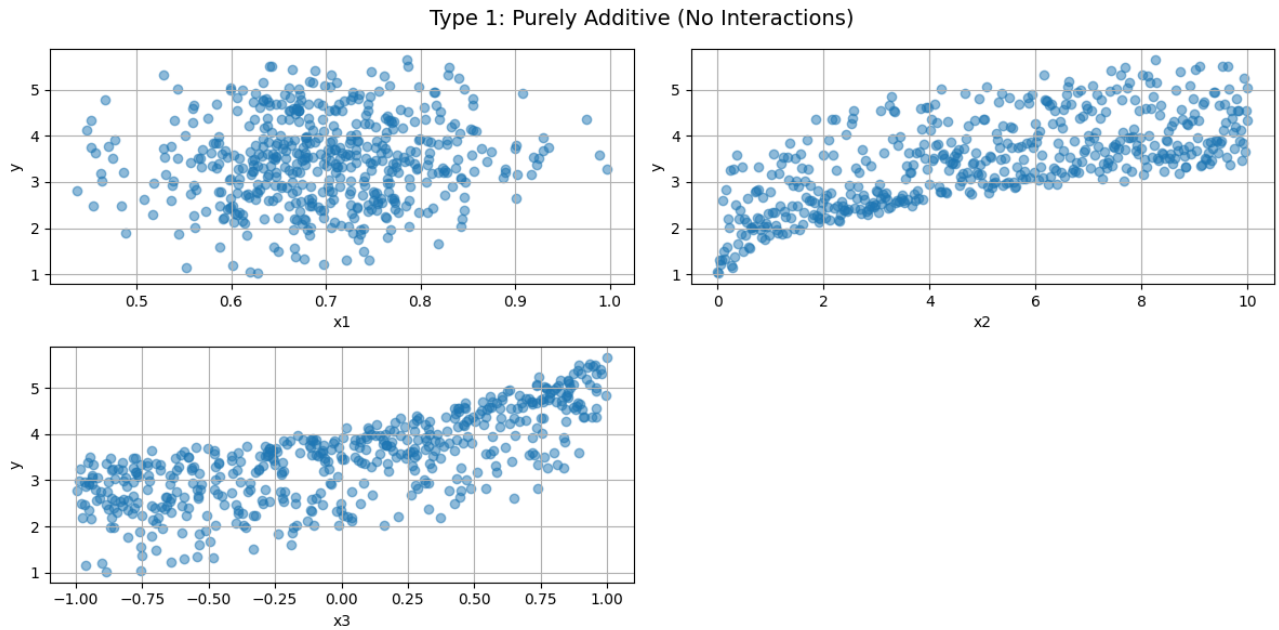


Figure 5: Joint distribution of Type 1 data.

## 2.3 Type 2: Strong Interaction Between Features

This dataset is defined by a strong interaction term:

```
x_1 = np.random.uniform(0, 2, 500)
x_2 = np.random.uniform(-1, 1, 500)
y = x_1 * x_2
```

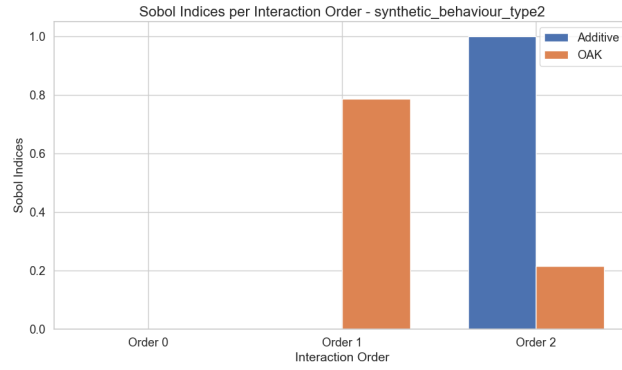


Figure 6: Sensitivity analysis for Type 2 data.

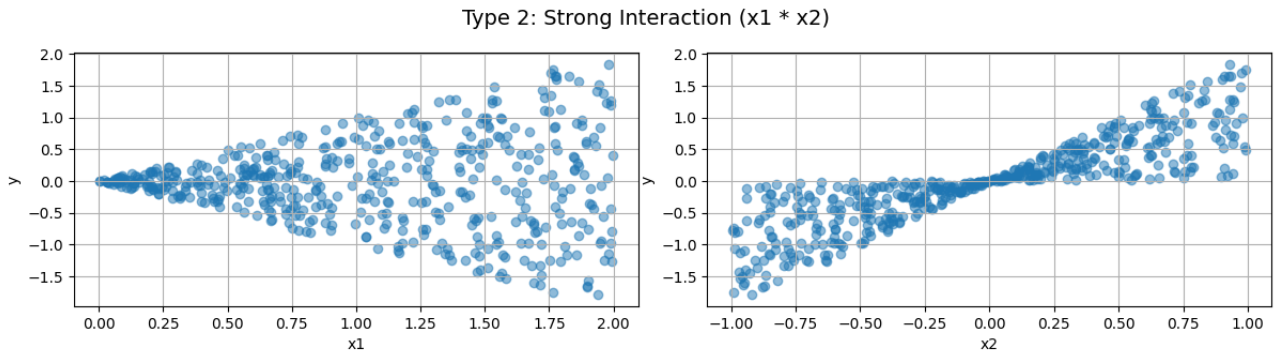


Figure 7: Marginal distributions of Type 2 data.

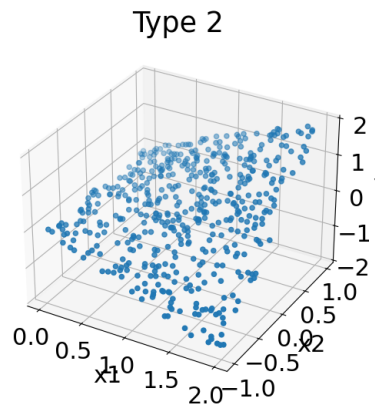


Figure 8: Joint distribution of Type 2 data.

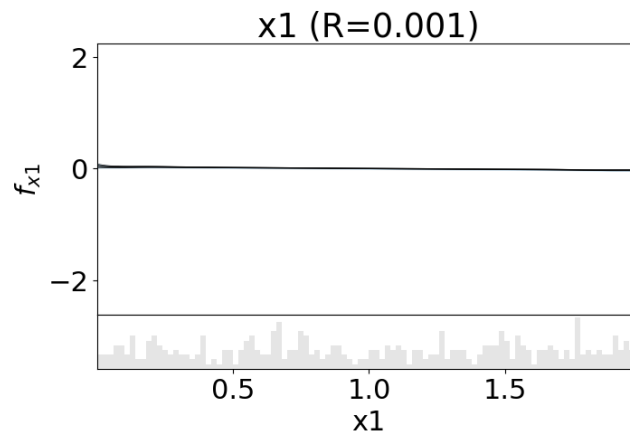


Figure 9: The fitted OAK function for  $x_1$

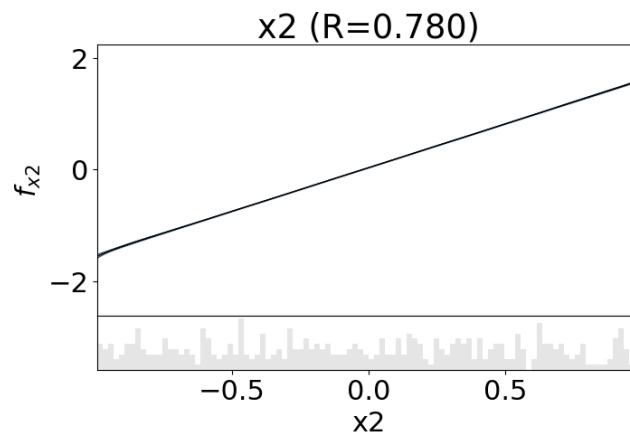


Figure 10: The fitted OAK function for  $x_2$

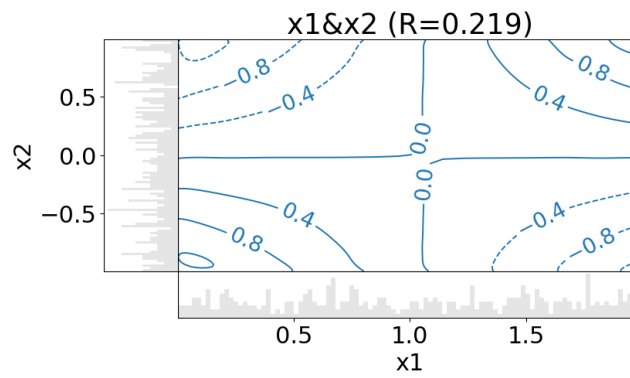


Figure 11: The fitted OAK function for interaction of  $x_1$ ,  $x_2$

## 2.4 Type 3: Sparse Dependency on Features

The response variable depends only on a subset of features:

```
x_1 = np.random.uniform(-2, 2, 500)
x_2 = np.random.normal(0, 1, 500)
x_3 = np.random.uniform(0.5, 2, 500)
y = np.sin(x_1) + np.log(1 + x_3)
```

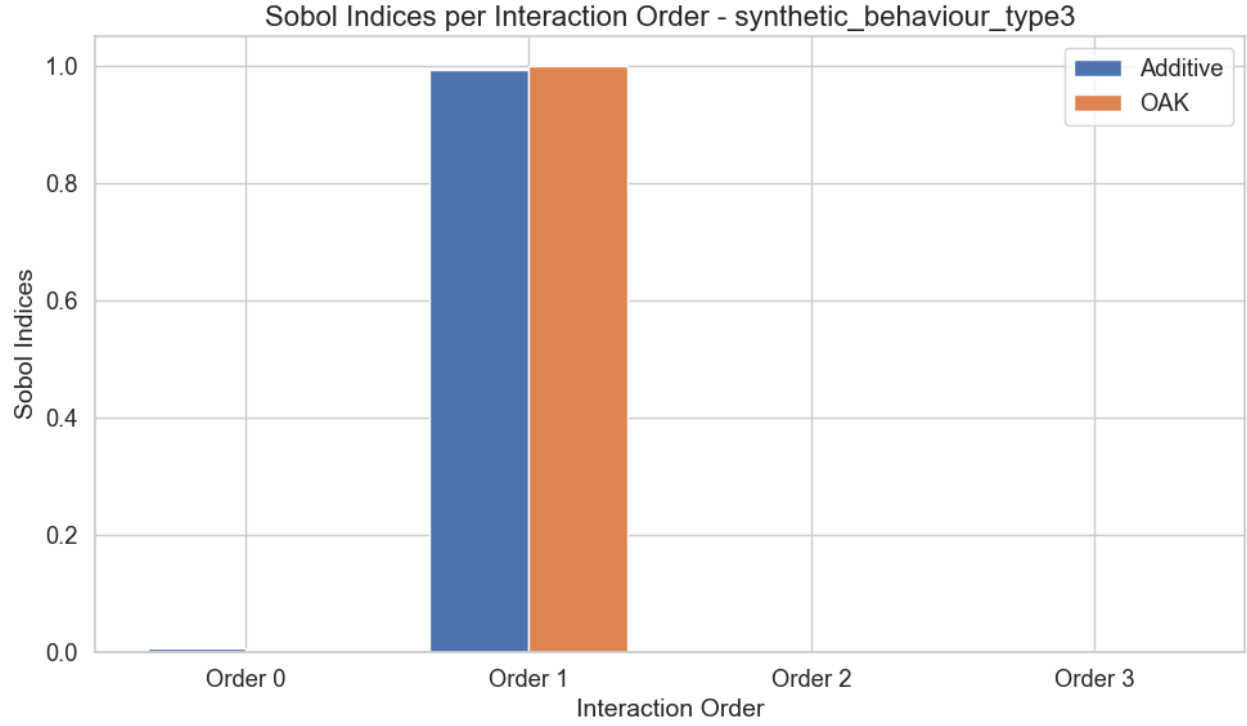


Figure 12: Sensitivity analysis for Type 3 data.

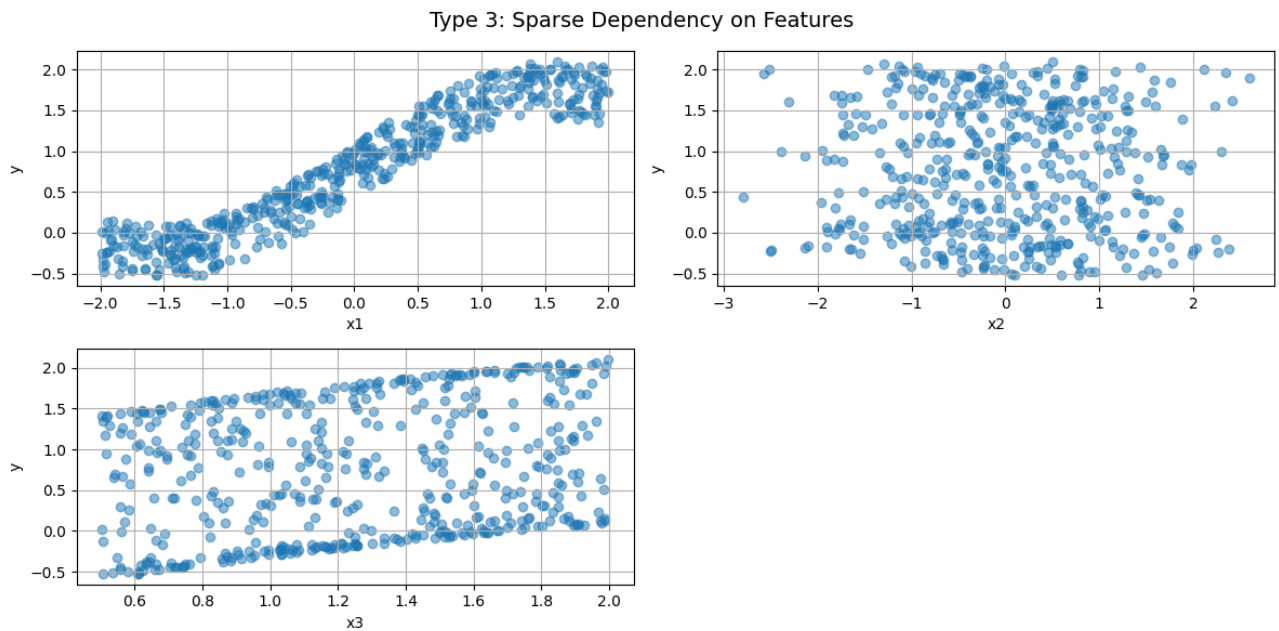


Figure 13: Marginal distributions of Type 3 data.

## 2.5 Type 4: Low dimensional Dependency on Features

The response variable depends only on a subset of features:

```
x_1 = np.linspace(-2, 2, 500)
x_2 = np.random.uniform(-1, 1, 500)
x_3 = np.random.uniform(0, 1, 500)
y = np.sin(x_1) * np.cos(x_2) + np.exp(-x_3)
```

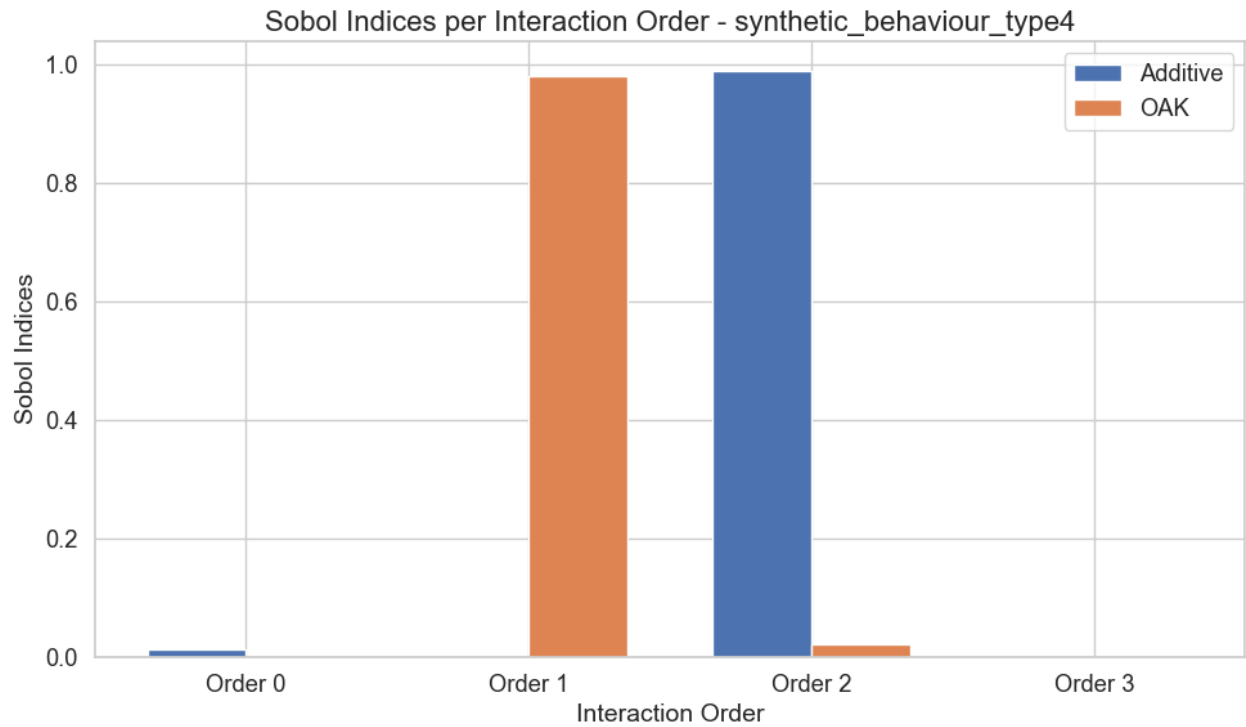


Figure 14: Sensitivity analysis for Type 4 data.

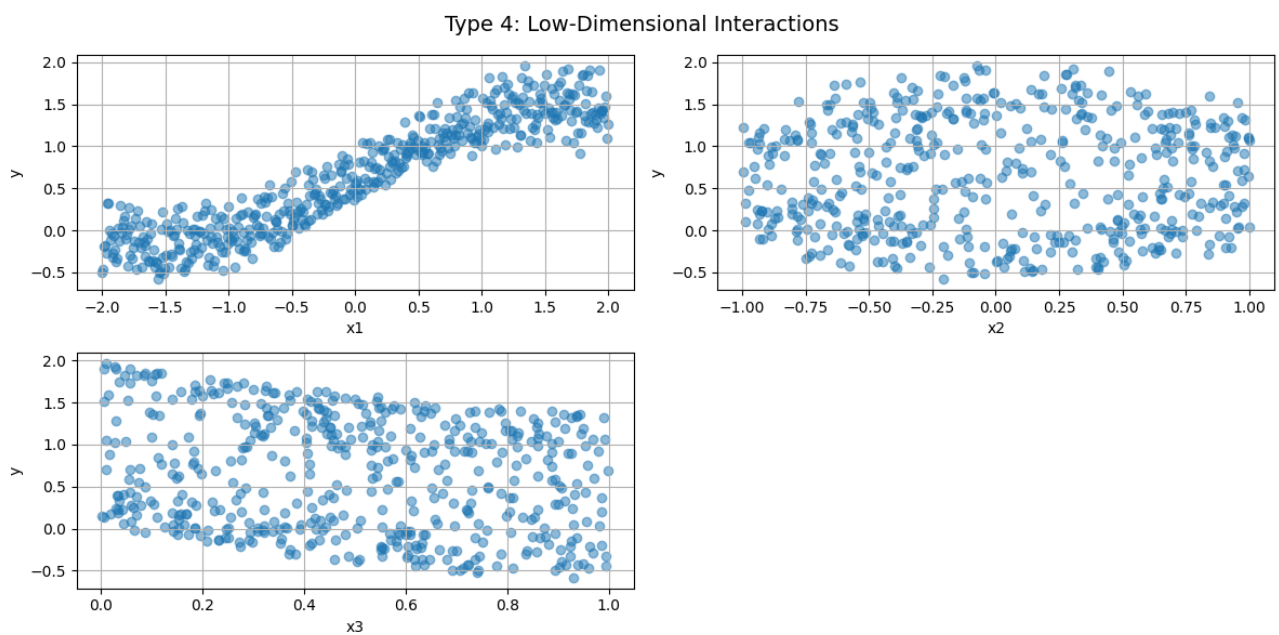


Figure 15: Marginal distributions of Type 3 data.



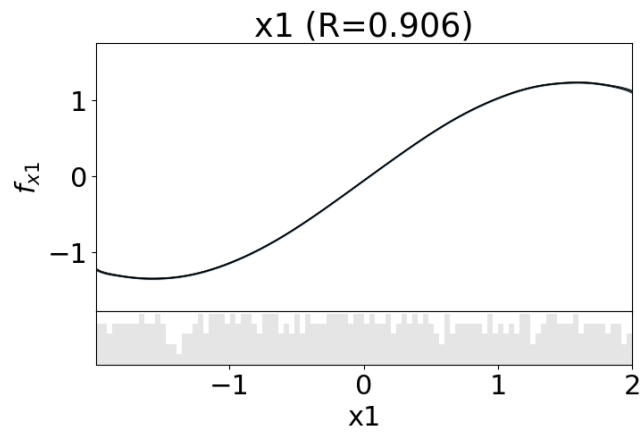


Figure 16: Enter Caption

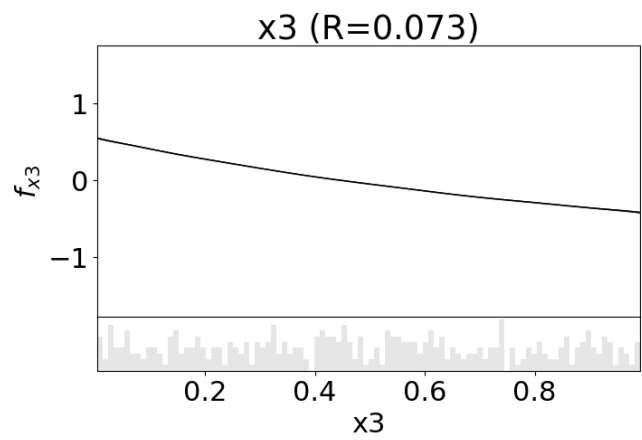


Figure 17: Enter Caption

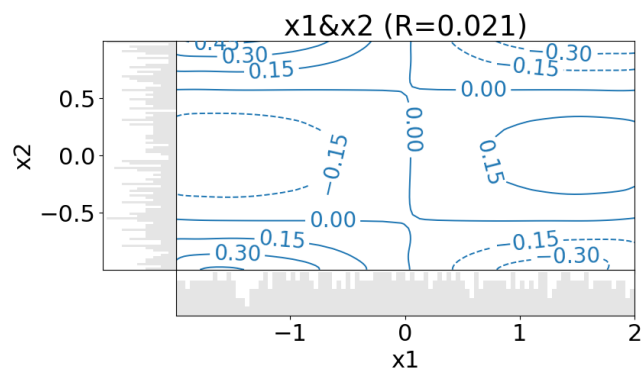


Figure 18: Enter Caption

## 2.6 Type 5: Mixed Terms

The response variable depends only on a subset of features:

```
x_1 = np.random.normal(0, 1, 500)
x_2 = np.random.normal(0, 1, 500)
x_3 = np.random.uniform(0, 2, 500)
x_4 = np.random.uniform(-1, 1, 500)
y = x_1**2 + x_2**2 + x_3 * x_4
```

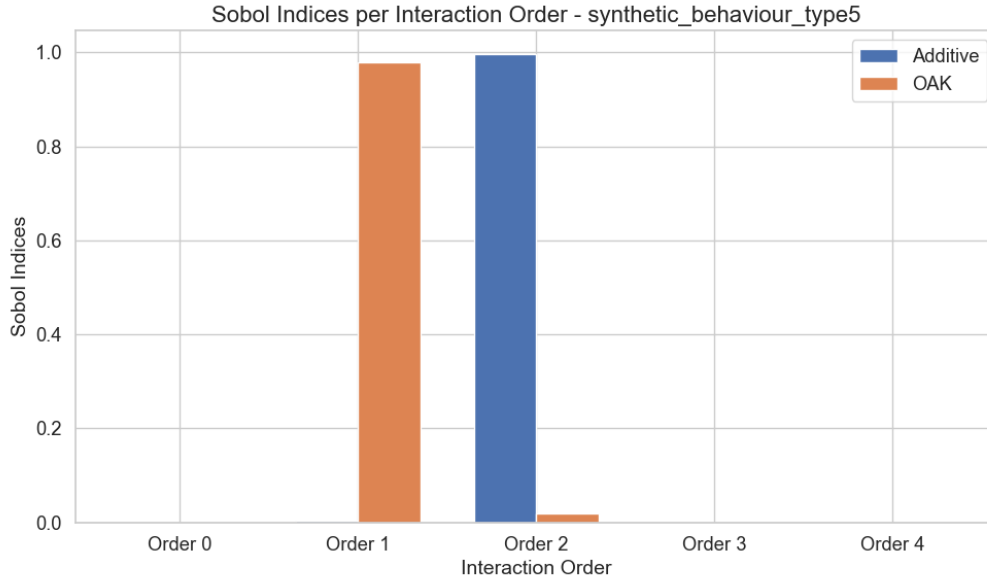


Figure 19: Sensitivity analysis for Type 5 data.

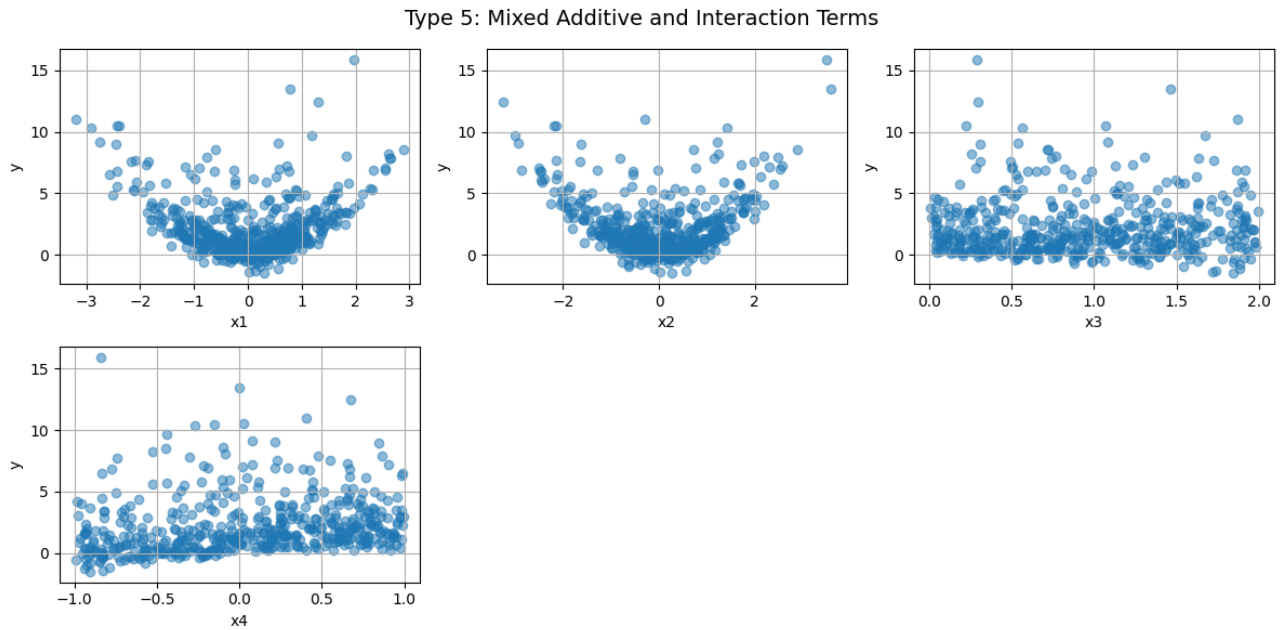


Figure 20: Marginal distributions of Type 5 data.

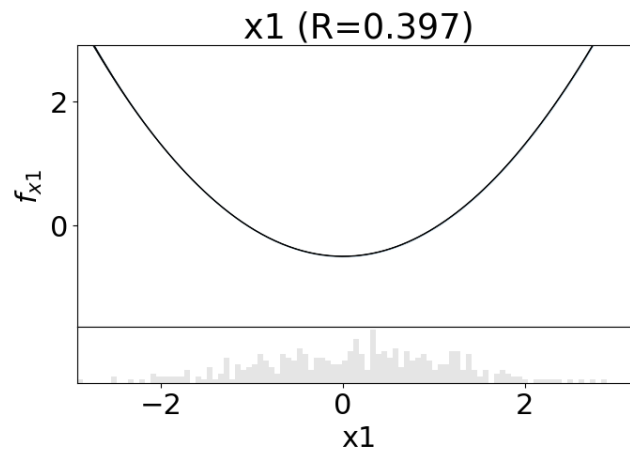


Figure 21: Enter Caption

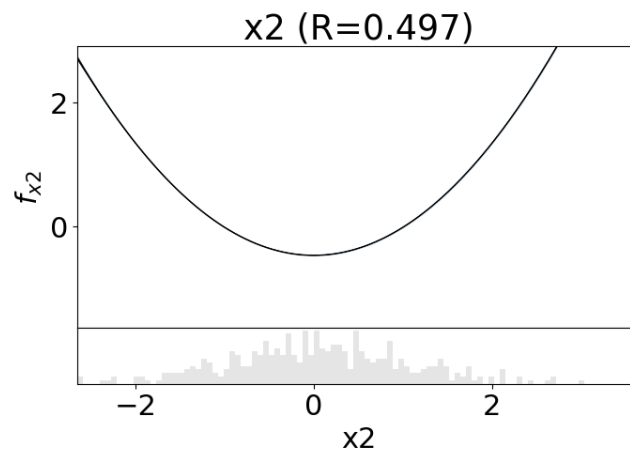


Figure 22: Enter Caption

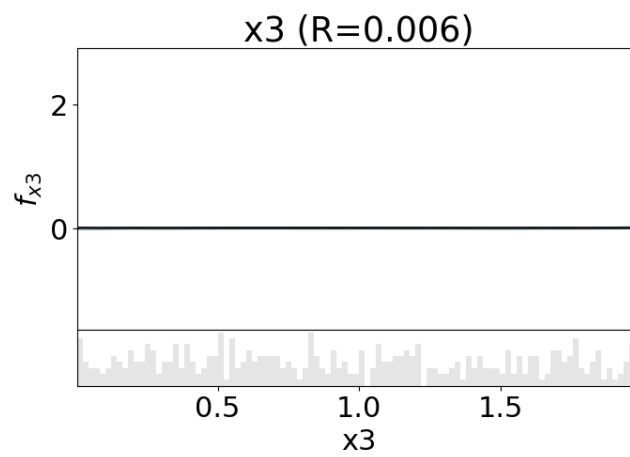


Figure 23: Enter Caption

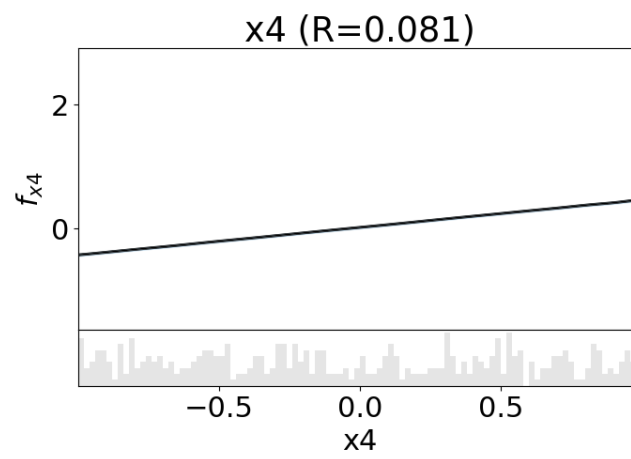


Figure 24: Enter Caption

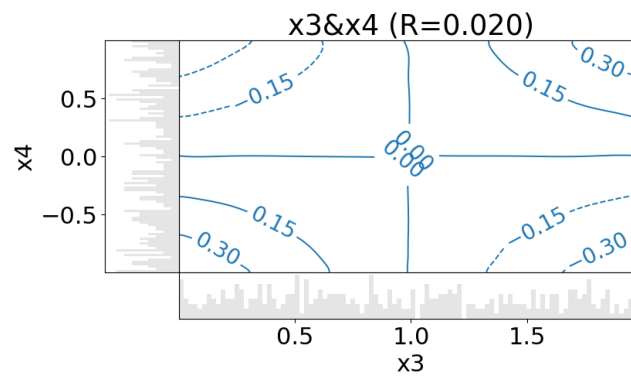


Figure 25: Enter Caption

### 3 AutoMPG Dataset

In addition to the synthetic datasets, we include an analysis of the AutoMPG dataset.

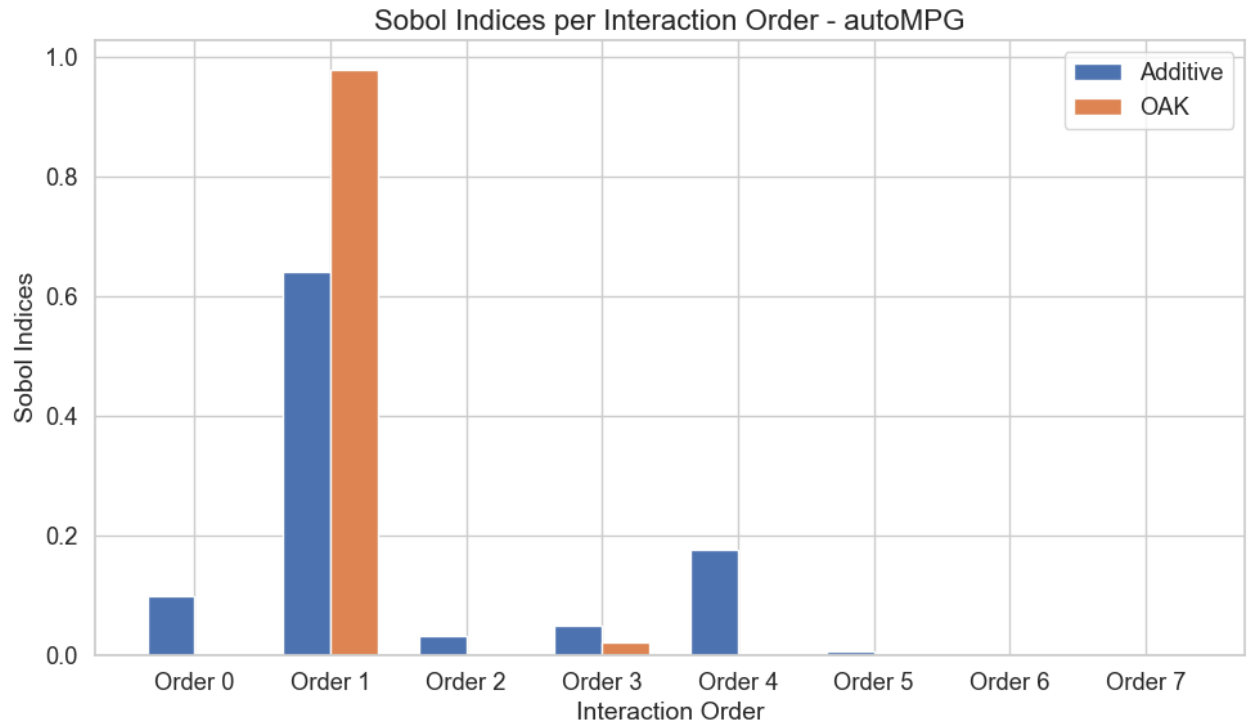


Figure 26: Sensitivity analysis for AutoMPG dataset.

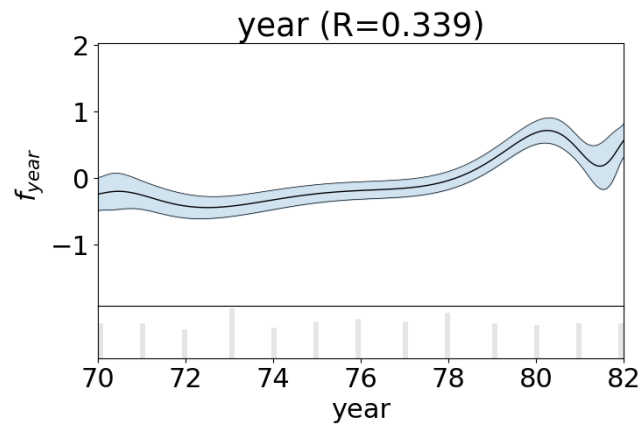


Figure 27: Enter Caption

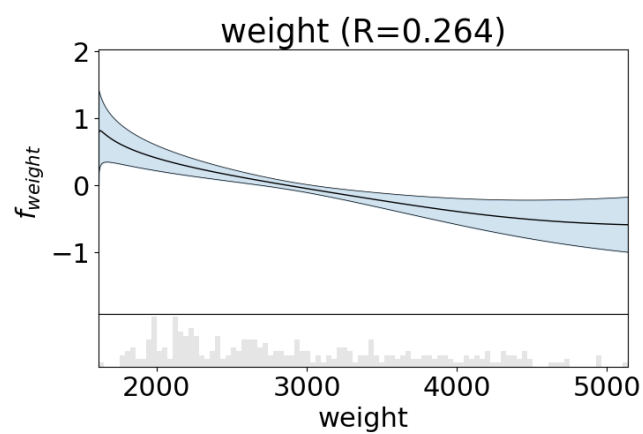


Figure 28: Enter Caption

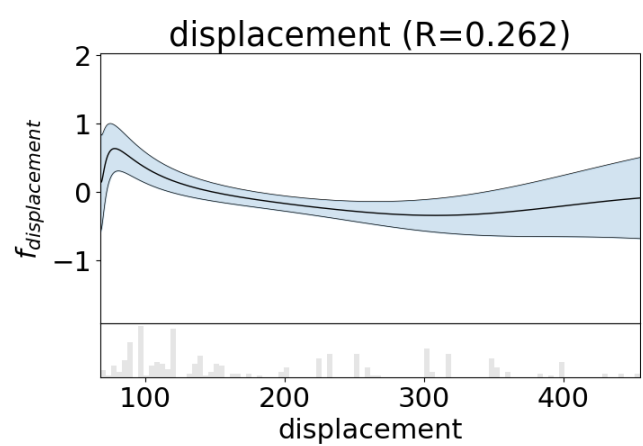


Figure 29: Enter Caption

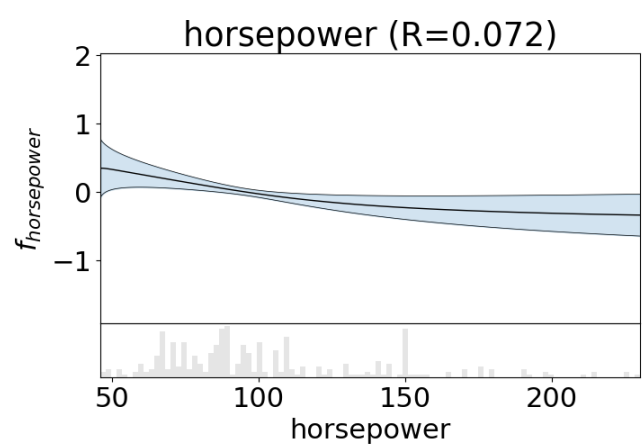


Figure 30: Enter Caption

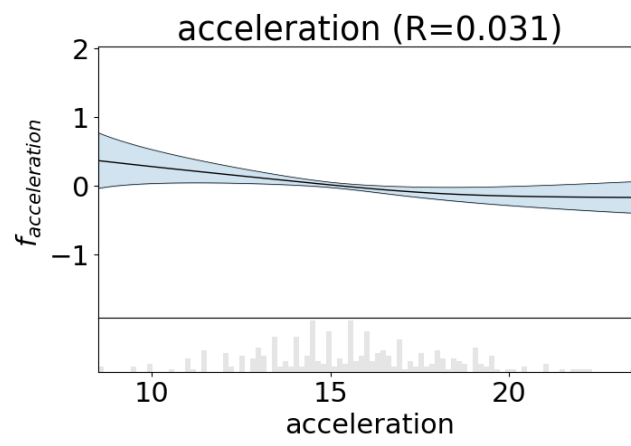


Figure 31: Enter Caption